

Q1 (20 July 2021 Shift 1)

A circular disc reaches from top to bottom of an inclined plane of length 'L'. When it slips down the plane, it takes time ' t_1 '. When it rolls down the plane, it takes time t_2 . The value of $\frac{t_2}{t_1}$ is $\sqrt{\frac{3}{x}}$. The value of x will be _____

Q2 (20 July 2021 Shift 2)

A body rolls down an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is :

- (1) Solid sphere
- (2) Solid cylinder
- (3) Hollow cylinder
- (4) Ring

Q3 (20 July 2021 Shift 2)

Two bodies, a ring and a solid cylinder of same material are rolling down without slipping an inclined plane. The radii of the bodies are same. The ratio of velocity of the centre of mass at the bottom of the inclined plane of the ring to that of the cylinder is $\frac{\sqrt{x}}{2}$. Then, the value of x is _____.

Q4 (20 July 2021 Shift 2)

A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800rpm in 10sec. The number of rotations made in the process is

Q5 (22 July 2021 Shift 1)

Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter. The correct statement for this situation is:-

- (1) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.
- (2) The ring has the greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane.
- (3) All of them will have same velocity.
- (4) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

Q6 (22 July 2021 Shift 1)

6. The centre of a wheel rolling on a plane surface moves with a speed v_0 . A particle on the rim of the wheel at the same level as the centre will be moving at a speed $\sqrt{x}v_0$. Then the value of x is _____.

Q7 (25 July 2021 Shift 1)

Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**. Assertion **A** : Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x , I_y and I_z respectively. The respective radii of gyration about all the three axes will be the same.

Reason **R** : A rigid body making rotational motion has fixed mass and shape. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
- (2) **A** is not correct but **R** is correct.
- (3) **A** is correct but **R** is not correct.
- (4) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

Q8 (25 July 2021 Shift 1)

A particle of mass 'm' is moving in time 't' on a trajectory given by

$$\vec{r} = 10\alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}$$

Where α and β are dimensional constants.

The angular momentum of the particle becomes the same as it was for $t = 0$ at time $t =$ _____ seconds.

Q9 (25 July 2021 Shift 2)

A solid disc of radius 20 cm and mass 10 kg is rotating with an angular velocity of 600rpm, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10 s is

$$\pi \times 10^{-1} \text{ Nm}$$

Q10 (27 July 2021 Shift 1)

List-I	List-II
(a) MI of the rod (length L, Mass M, about an axis $\perp$ to the rod passing through the midpoint)	(i) $8ML^2/3$
(b) MI of the rod (length L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(ii) $ML^2/3$
(c) MI of the rod (length 2 L, Mass M, about an axis $\perp$ to the rod passing through its midpoint)	(iii) $ML^2/12$
(d) MI of the rod (Length 2L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(iv) $2ML^2/3$

Choose the correct answer from the options given below :

(1) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(2) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)

(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

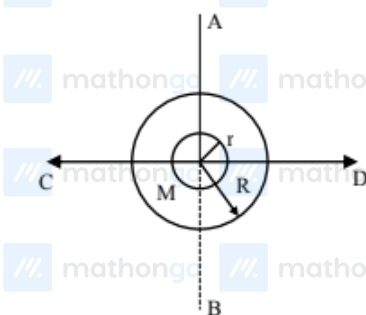
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q11 (27 July 2021 Shift 1)

The figure shows two solid discs with radius R and r respectively. If mass per unit area is same for both, what is the ratio of

MI of bigger disc around axis AB (Which is $\perp$ to the plane of the disc and passing through its centre) of MI of smaller disc around one of its diameters lying on its plane?

Given 'M' is the mass of the larger disc. (MI stands for moment of inertia)



(1) $R^2 : r^2$

(2) $2r^4 : R^4$

(3) $2R^2 : r^2$

(4) $2R^4 : r^4$

Q12 (27 July 2021 Shift 2)

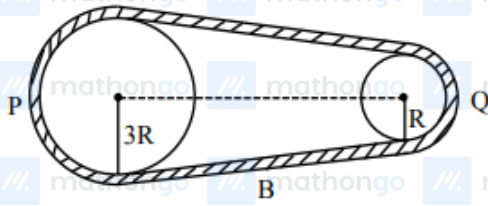
In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational

Questions with Answer Keys

MathonGo

kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$

will be $x : 1$. The value of x will be



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Answer Key

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Q1 (2)

Q2 (2)

Q3 (3)

Q4 (200)

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Q5 (1)

Q6 (2)

Q7 (2)

Q8 (10)

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Q9 (4)

Q10 (3)

Q11 (4)

Q12 (9)

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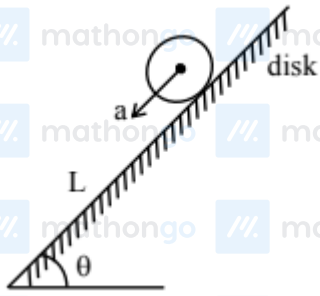
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Q1



If disk slips on inclined plane, then its acceleration

$$a_1 = g \sin \theta$$

$$L = \frac{1}{2} a_1 t_1^2$$

$$\Rightarrow t_1 = \sqrt{\frac{2L}{a_1}} \dots (i)$$

If disk rolls on inclined plane, its acceleration,

$$a_2 = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$a_2 = \frac{g \sin \theta}{1 + \frac{mR^2}{2mR^2}}$$

$$a_2 = \frac{2}{3} g \sin \theta$$

$$\text{Now } L = \frac{1}{2} a_2 \cdot t_2^2$$

$$\Rightarrow t_2 = \sqrt{\frac{2L}{a_2}} \dots (ii)$$

$$\text{Now } \frac{t_2}{t_1} = \sqrt{\frac{a_1}{a_2}} = \sqrt{\frac{3}{2}}$$

$$\Rightarrow x = 2$$

Q2

$$\frac{1}{2} I \omega^2 = \frac{1}{2} \times \frac{1}{2} m v^2$$

$$I = \frac{1}{2} m R^2$$

Body is solid cylinder

Q3

Hints and Solutions

MathonGo

I in both cases is about point of contact

Ring

$$mgh = \frac{1}{2} I \omega^2$$

$$mgh = \frac{1}{2} (2mR^2) \frac{v_R^2}{R^2}$$

$$v_R = \sqrt{gh}$$

Solid cylinder

$$mgh = \frac{1}{2} I \omega^2$$

$$mgh = \frac{1}{2} \left(\frac{3}{2} mR^2 \right) \frac{v_C^2}{R^2}$$

$$v_C = \sqrt{\frac{4gh}{3}}$$

$$\frac{v_R}{v_C} = \frac{\sqrt{3}}{2}$$

Q4

$$\omega_f = \omega_0 + \alpha t$$

$$\alpha = 1200 \times 6$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$= 600 \times \frac{10}{60} + \frac{1}{2} \times 1200 \times 6 \times \frac{1}{36}$$

$$\theta = 200$$

Q5

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$I_{\text{ring}} > I_{\text{solid cylinder}} > I_{\text{solid sphere}}$$

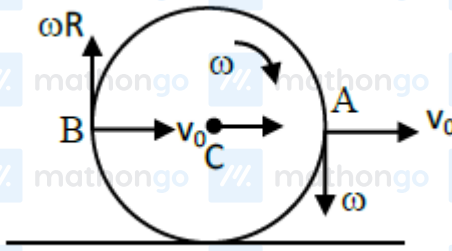
$$\Rightarrow a_{\text{ring}} < a_{\text{solid cylinder}} < a_{\text{solid sphere}}$$

$$\Rightarrow v_{\text{ring}} < v_{\text{solid cylinder}} < v_{\text{solid sphere}}$$

Q6

Hints and Solutions

MathonGo



For no slipping $v_0 = \omega R$

$$\text{Now } v_A = v_B = \sqrt{v_0^2 + (\omega R)^2}$$

$$= \sqrt{2}v_0$$

$$\Rightarrow x = 2$$

Q7

$I_z = I_x + I_y$ (using perpendicular axis theorem) & $I = mk^2$ (K : radius of gyration)

$$\text{so } mK_z^2 = mK_x^2 + mK_y^2$$

$$K_z^2 = K_x^2 + K_y^2$$

so radius of gyration about axes x, y & z won't be

same hence assertion A is not correct reason R is

correct statement (property of a rigid body)

Q8

$$\vec{r} = 10\alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}$$

$$\vec{v} = 20\alpha \hat{i} + 5\beta \hat{j}$$

$$\vec{L} = m(\vec{r} \times \vec{v})$$

$$= m [10\alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}] \times [20\alpha \hat{i} + 5\beta \hat{j}]$$

$$\vec{L} = m [50\alpha\beta t^2 \hat{k} - 100\alpha\beta(t-5)\hat{k}]$$

$$\text{At } t = 0, \vec{L} = \vec{0}$$

$$50\alpha\beta t^2 - 100\alpha\beta(t-5) = 0$$

Hints and Solutions

MathonGo

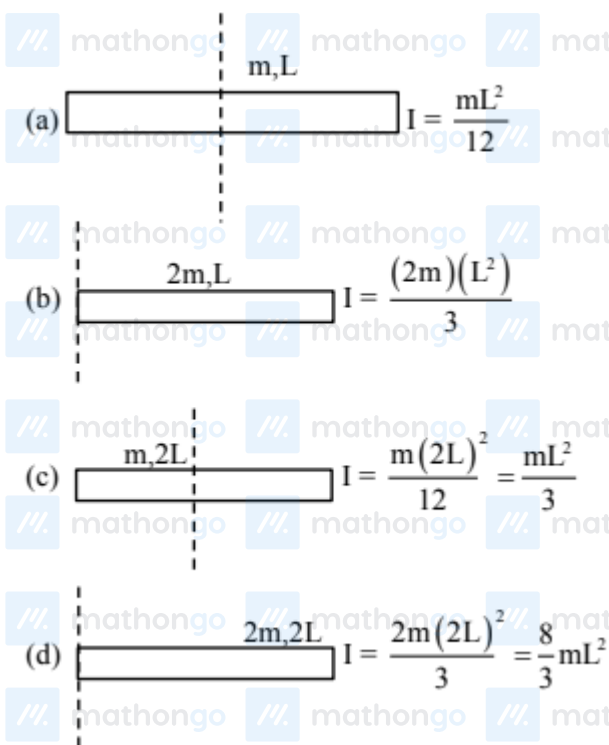
$$t - 2(t - 5) = 0$$

$$t = 10 \text{ sec}$$

Q9

$$\begin{aligned} \tau &= \frac{\Delta L}{\Delta t} = \frac{I(\omega_f - \omega_i)}{\Delta t} \\ \tau &= \frac{\frac{mR^2}{2} \times [0 - \omega]}{\Delta t} \\ &= \frac{10 \times (20 \times 10^{-2})^2}{2} \times \frac{600 \times \pi}{30 \times 10} \\ &= 0.4\pi = 4\pi \times 10^{-2} \end{aligned}$$

Q10



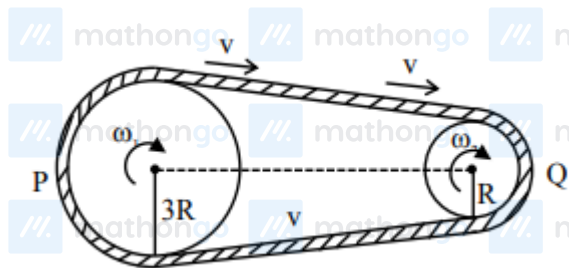
Q11

$$\begin{aligned} \text{Ratio of moment of inertia} &= \frac{\frac{1}{2}MR^2}{\frac{1}{4}mr^2} \\ &= \frac{2\sigma\pi R^2 R^2}{\sigma\pi r^2 r^2} = \frac{2R^4}{r^4} \end{aligned}$$

Q12

Hints and Solutions

MathonGo



$$\frac{1}{2} I_1 (\omega_1)^2 = \frac{1}{2} I_2 (\omega_2)^2$$

$$I_1 \left(\frac{v}{3R} \right)^2 = I_2 \left(\frac{v}{R} \right)^2$$

$$\frac{I_1}{I_2} = \frac{9}{1}$$